Status	Finished
Started	Sunday, 18 October 2026, 3:31 PM
Completed	Sunday, 18 October 2026, 3:45 PM
Duration	14 mins 30 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is very small
b. When the list contains duplicate elements
c. When the list is sorted ✓
d. When the list is unsorted

Question **2**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the search algorithm finishes
b. When the element is found in the list ✓
c. When the list is sorted
d. When the list contains duplicate elements

Question **3**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 89 and 99
b. 89 and 94
c. 90 and 99 ✓
d. 90 and 94

Question **4**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

- a. It can be applied only if the list is sorted ✓
- b. It compares elements sequentially
- c. It always starts from the beginning of the list
- d. It works on unsorted lists

Question **5**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

- a. Linear search ✓
- b. Both (1) & (3)
- c. Hash search
- d. Binary search

Question **6**

Correct

Mark 1.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

- a. The lower middle element is chosen as the middle element
- b. The search stops
- c. The middle element is chosen randomly
- d. The higher middle element is chosen as the middle element ✓

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Must use a sorted array

b. Binary search algorithm is not efficient when the data elements more than 1500 ✓

c. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

d. There must be a mechanism to access middle element directly

Question **8**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Linear search ✓

b. Binary search

c. Hash search

d. Interpolation search

Question **9**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

a. Compare the target element with the middle element in the list ✓

b. Divide the list into two equal parts

c. Compare the target element with the first element in the list

d. Sort the list

Question **10**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Sequential search

b. Hash search

c. Binary search ✓

d. Linear search

Question **11**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By using a hash function
b. By comparing each element in the list sequentially ✓
c. By dividing the list into halves
d. By sorting the list first

Question **12**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By starting from the first element
b. By comparing each element sequentially
c. By using a hash function
d. By dividing the list length by two ✓

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. The list must be sorted
b. Number values should only be present

Number values should only be present

c. There must be a mechanism to delete and/or insert elements in the list ✓
d. There should be direct access to the middle element in any sublist

Question **14**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is small and sorted
b. When the list is small and unsorted ✓
c. When the list is large and unsorted
d. When the list is large and sorted

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It can only be applied to large lists
b. It requires the list to be sorted ✓
c. It is slower than linear search for small lists
d. It does not work with negative numbers